

PYTHON PROGRAMMING

Assignment: Comprehensions in Python

List • Dictionary • Nested • Mixed

Total Marks: 50 | Time: 45 minutes | All questions are compulsory

Instructions

1. Write your answers in the space provided below each question.
2. For coding questions, write the complete Python code including variable names.
3. Marks are indicated alongside each question.
4. You may use any valid comprehension syntax unless stated otherwise.

SECTION A — List Comprehensions (20 marks)

Q1. Cubes & Squares List

Easy 5 pts

Using list comprehensions, create TWO separate lists: (a) A list of CUBES for numbers 1 through 10. (b) A list of SQUARES for numbers 1 through 10. Print both lists.

Starter code:

```
# (a) Cubes

cubes = [ _____ for x in range(1, 11) ]

# (b) Squares

squares = [ _____ for x in range(1, 11) ]
```

*Hint: $x**3$ gives the cube of x . Use `range(1, 11)` as the iterable.*

Answer / Output:

Q2. Uppercase Converter

Easy 5 pts

You are given a list of product names in lowercase. Using a list comprehension, convert every product name to UPPERCASE. Print the resulting list.

Starter code:

```
products = ["laptop", "mouse", "keyboard", "monitor", "webcam"]

# Your list comprehension here

upper_products = [ _____ ]

print(upper_products)
```

Hint: Strings have an .upper() method.

Answer / Output:

Q3. GST Calculator

Easy 5 pts

A shopkeeper has a list of product prices (before tax). India's GST rate is 18%. Using a list comprehension, generate a new list containing the final price of each product AFTER adding 18% GST. Round each value to 2 decimal places.

Starter code:

```
prices = [499, 1299, 249, 3999, 149, 799]

gst_prices = [ _____ for price in prices ]

print(gst_prices)
```

*Hint: Final price = price * 1.18. Use round(value, 2) to round.*

Answer / Output:

Q4. Name Filter

Easy 5 pts

Given a list of student names, use a list comprehension to create a new list that contains ONLY those names that start with the letter 'A' or 'J' (case-sensitive). Print the filtered list.

Starter code:

```
students = ["Ronak", "Ananya", "Karan", "Jaya", "Mehul",  
            "Arjun", "Priya", "Jasmin", "Dhruv", "Aditya"]  
  
result = [ name for name in students if _____ ]  
  
print(result)
```

Hint: Use `name[0] == "A"` or `name.startswith("J")` as the condition.

Answer / Output:

SECTION B — Dictionary Comprehensions (15 marks)

Q5. Squares Dictionary

Easy 5 pts

Create a dictionary comprehension that maps each integer from 1 to 5 to its SQUARE. The expected output is: { 1: 1, 2: 4, 3: 9, 4: 16, 5: 25 }

Starter code:

```
squares_dict = { _____ : _____ for i in range(1, 6) }  
print(squares_dict)
```

Hint: Syntax: {key: value for item in iterable}

Answer / Output:

Q6. Salary Hike

Medium 5 pts

The HR department wants to give every employee a 15% salary raise. Using a dictionary comprehension, create an updated salary dictionary. Print both the original and updated dictionaries.

Starter code:

```
employees = {  
    "Alice": 45000,  
    "Bob": 52000,  
    "Charlie": 38000,  
    "Diana": 61000,  
}  
  
updated = { _____ : _____ for name, salary in employees.items() }  
print(updated)
```

Hint: Iterate over employees.items() to get (name, salary) pairs.

Answer / Output:

Q7. High Earners Filter

Medium 5 pts

Using a dictionary comprehension WITH a condition, create a new dictionary that contains ONLY those employees whose salary is GREATER THAN 50,000. Use the same employees dictionary from Q6.

Starter code:

```
# Use the employees dict from Q6

high_earners = { _____ for name, salary in employees.items()
                 _____ }

print(high_earners)
```

Hint: Add 'if salary > 50000' after the for clause.

Answer / Output:

SECTION C — Nested & Mixed Comprehensions (15 marks)

Q8. Flatten & Filter Matrix

Medium 5 pts

You are given the matrix below. Using a SINGLE nested list comprehension: (a) Flatten the matrix into a one-dimensional list. (b) Include ONLY even numbers. Expected output: [2, 4, 6, 8]

Starter code:

```
matrix = [  
    [1, 2, 3],  
    [4, 5, 6],  
    [7, 8, 9],  
]  
  
result = [ _____ for row in matrix for cell in row if _____ ]  
  
print(result)
```

Hint: Use two 'for' clauses inside the comprehension. The condition goes at the end.

Answer / Output:

Q9. Multiplication Table

Medium 5 pts

Using a nested list comprehension, generate a 5x5 multiplication table. The result should be a list of 5 lists, where `table[i][j] = (i+1) * (j+1)`. Print the table row by row.

Starter code:

```
table = [ [_____] for i in range(1, 6) ]

for row in table:

    print(row)
```

Hint: Outer loop: `i in range(1,6)`. Inner loop: `j in range(1,6)`. Expression: `i*j`.

Answer / Output:

Q10. Bonus Challenge — Classify Numbers

Hard 5 pts

Using a SINGLE list comprehension with a ternary (if-else) expression, create a list that classifies numbers 1 through 15: - If divisible by 3 AND 5 → 'FizzBuzz' - If divisible by 3 only → 'Fizz' - If divisible by 5 only → 'Buzz' - Otherwise → the number itself (as a string) Expected: [1,'2','Fizz','4','Buzz','Fizz','7','8','Fizz','Buzz','11','Fizz','13','14','FizzBuzz']

Starter code:

```
result = [  
    'FizzBuzz' if _____ else  
    'Fizz' if _____ else  
    'Buzz' if _____ else  
    str(n)  
    for n in range(1, 16)  
]  
print(result)
```

Hint: You can chain ternary expressions: val1 if cond1 else val2 if cond2 else val3 if cond3 else val4

Answer / Output:

Score Sheet

Question	Topic	Max	Obtained
Q1	Cubes & Squares — List Comprehension	5	
Q2	Uppercase Converter — List Comprehension	5	
Q3	GST Calculator — List Comprehension	5	
Q4	Name Filter — Conditional List Comprehension	5	
Q5	Squares Dictionary — Dict Comprehension	5	
Q6	Salary Hike — Dict Comprehension	5	
Q7	High Earners Filter — Dict Comprehension with Condition	5	
Q8	Flatten & Filter Matrix — Nested Comprehension	5	
Q9	Multiplication Table — Nested Comprehension	5	
Q10	FizzBuzz — Mixed / Bonus	5	
	TOTAL	50	

Student Declaration

Name:	_____
Roll No.:	_____
Date:	_____
Signature:	_____